

Is My Text in Your AI Model?

Gradient-based Membership Inference Test applied to LLMs

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Abstract—This work adapts and studies the gradient-based Membership Inference Test (gMINT) to the classification of text based on LLMs. MINT is a general approach intended to determine if given data was used for training machine learning models, and this work focuses on its application to the domain of Natural Language Processing. Using gradient-based analysis, the MINT model identifies whether particular data samples were included during the language model training phase, addressing growing concerns about data privacy in machine learning. The method was evaluated in seven Transformer-based models and six datasets comprising over 2.5 million sentences, focusing on text classification tasks. Experimental results demonstrate MINT's robustness, achieving AUC scores between 85% and 99%, depending on data size and model architecture. These findings highlight MINT's potential as a scalable and reliable tool for auditing machine learning models, ensuring transparency, safeguarding sensitive data, and fostering ethical compliance in the deployment of AI/NLP technologies.

I. INTRODUCTION

In recent years, Natural Language Processing (NLP) technologies have been increasingly deployed in various sectors such as healthcare [1], legal [2], [3], customer service [4], education [5], [6], or finance [7], profoundly changing the way information is analyzed and understood. Advancements in machine learning have enabled the processing of vast textual data, driving rapid progress in NLP. However, these developments have also raised important ethical and legal concerns that require immediate attention [8]. In response, the European Union introduced new legislation in June 2024¹, specifically aimed at regulating the use of AI technologies. The proposed legislation represents a unified effort to address the ethical and social implications associated with the widespread use of AI in general and NLP systems in particular. Its primary goal is to protect the fundamental rights of individuals through measures that ensure transparency and accountability in the development and implementation of AI technologies. That legal framework and related international guidelines enforce the creation of audit tools to oversee the responsible integration of AI/NLP systems into society, while also gaining a deeper understanding of their functioning and associated risks [9].

In this scenario of raising privacy concerns in AI deployments [10], we focus our attention on Membership Inference

Attacks (MIAs), which involve adversaries attempting to extract sensitive information about the data used to train a model [11]. Research has demonstrated the feasibility of these attacks and has proposed various strategies to address them. Building on this foundation, [12] introduced an innovative framework called the Membership Inference Test (MINT), designed to detect unauthorized data use when training AI/ML models, which was soon followed by a gradient-based variant, gMINT [13]. The primary objective of MINT approaches (including gMINT) is to determine whether specific data have been used without the necessary permissions. (A MINT demonstrator was recently released by [14].²) This approach seeks to audit AI and machine learning models, improving transparency and explainability [15], [16]. It enables individuals to identify whether certain data, such as personal information or other sensitive data [17], [18], were used as training data. Although MIAs and MINT share similar objectives, they operate in different contexts and address different aspects of the problem.

In line with recent research oriented to develop new technology for auditing LLMs [19], [20], the present paper addresses the challenge of identifying whether a given text was utilized in the training of a Language Model (see Fig. 1). The study seeks to further emphasize the versatility and applicability of the general MINT approach, initially validated for images by [12], by demonstrating its effectiveness in language-related tasks. Specifically, we adapt MINT to handle the unique characteristics of text data and language models. The key novel contributions are as follows:

- We adjust and apply MINT for the first time to LLMs, in particular the gradient-based variant (gMINT) proposed by [13]. This adaptation involves modifying how MINT processes and interprets textual data, ensuring that it captures the specific patterns and dependencies inherent in text. Using the gradients generated during the learning process, the gMINT model takes advantage of the patterns related to how each training sample influences the model's training and optimization.
- We evaluate gMINT in the context of text classification. This includes experimenting with seven Transformer-based Language Models and six datasets comprising 2.5 million sentences spanning diverse topics. The adapta-

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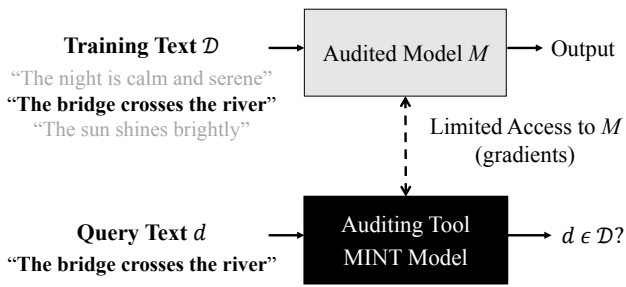


Fig. 1. The objective of MINT is to discern whether given data (d) were utilized in the training process of an AI Model (M) trained with a specific database ($\mathcal{D}$).

tion to text classification required fine-tuning the approach to account for the textual nature of the input data and the differences in model architectures compared to image-based models.

- The findings demonstrate that gMINT effectively identifies which samples belong to the training process with high accuracy, showcasing its ability to accurately trace the data used for model development in the text domain.

II. RELATED WORKS

A. Membership Inference Attacks (MIAs) in LLMs

Membership Inference Attacks (MIAs) aim to determine whether specific data samples were included in the training set of a machine learning model. These attacks exploit subtle behavioral differences in how a model processes training data compared to unseen data, such as variations in loss trajectories [21], similarity between distributions [22], or activation patterns [23]. However, executing MIAs is inherently challenging because most of the samples encountered by a model have not been used during training. This creates an imbalance between the data that the model has seen and the large amount of unseen data, making it difficult for an attacker to reliably infer membership [24].

For Language Models, MIAs face unique challenges due to the large volume and diversity of text data used during training, as well as the overlapping distributions between training and unseen data [25]. Existing methods, such as those proposed by [26], leverage loss comparisons with similar sentences generated via masked language models. Other approaches, such as user inference techniques, compare the likelihood of user-specific samples to determine membership [27]. Although effective, these methods are computationally expensive and often require extensive auxiliary data or shadow models, limiting their scalability. Furthermore, the scale and single-epoch nature of the language model pre-training reduce the visibility of training data, complicating membership inference [28]. The complexity of the language also creates challenges for attackers trying to identify discernible patterns. These factors make MIAs in language models especially complex, requiring advanced techniques to detect vulnerabilities while preserving the effectiveness and security of the model [29].

B. Membership Inference Test (MINT)

MINT is a Membership Inference Test designed to assess the compliance of machine learning models with current regulatory frameworks governing the use of training data. The process requires a certain level of cooperation from the model owner to ensure that the model's development and operation are aligned with legal and ethical guidelines. The central goal of MINT is to create robust methodologies that are applicable to various data modalities, such as text, images, and audio, while accounting for different levels of transparency and information shared by the model developer regarding the model's training and operational procedures [12]. These scenarios range from situations where little or no information is available to cases where the details of the model are fully disclosed [30].

Within the MINT framework, auditors are tasked with evaluating the model, having access either to the original model or to limited details about its structure and training processes. This aligns with the practical application of MINT, particularly in responding to evolving legal requirements and advancing data protection measures. Importantly, MINT distinguishes itself from traditional Membership Inference Attacks (MIAs), which typically involve creating shadow models to mimic the behavior of the original model. In contrast, MINT enables auditors to apply inference techniques directly to the original model, leading to different techniques and more accurate detection results. Several factors, such as the level of access to the model or the type of data modality, can influence the effectiveness and results of MINT, making it adaptable to a variety of compliance challenges [30].

To the best of our knowledge, MINT has not been studied in the context of language models. So far, it has only been studied in image intelligence settings [31], [32]. MINT can be an important auditing tool to enhance the transparency and trustworthiness of current and future language models.

III. MEMBERSHIP INFERENCE TEST: GRADIENT-BASED APPROACH (GMINT)

The original MINT proposal was tested on Face Recognition models. Here we study and develop MINT for Language Models, specifically models trained for text classification. These models exhibit several substantial differences with respect to image models: while image models typically output an embedding in a representation space, classification-oriented language models usually produce a probability distribution over a set of predefined classes. Both types of models rely on internal embeddings to represent their inputs, but they differ in how these embeddings are used: image models often expose the embedding directly as output, while language models use it to compute class probabilities.

Gradients can play a crucial role in MINT. When a data sample belongs to the training set, the model may react to it in a specific way, particularly reflected in the gradients, which in general will have distinct characteristics compared to gradients associated with data outside the training set. This behavior was leveraged by [13] generating the so-called gradient-based MINT (gMINT).

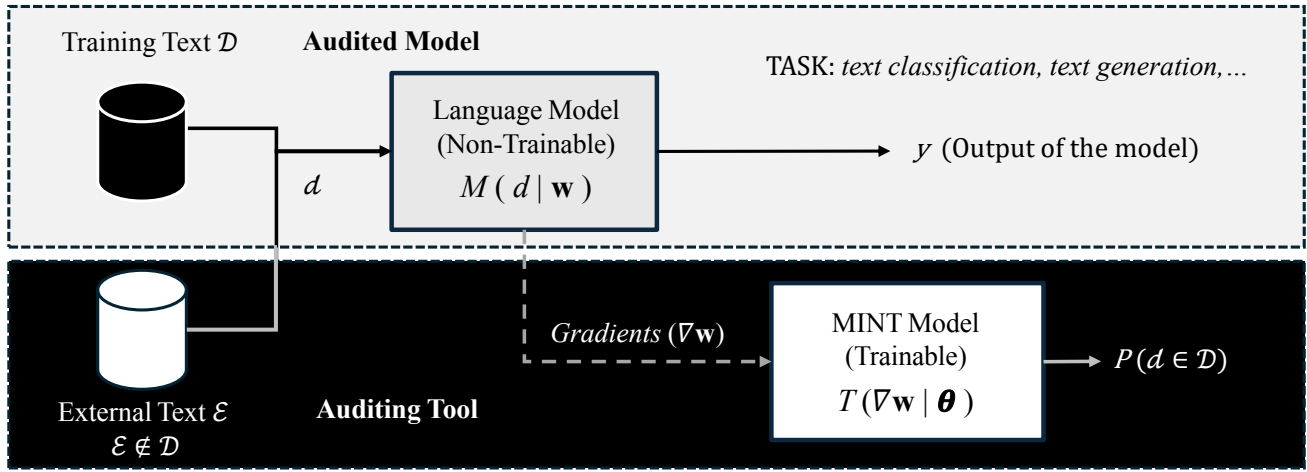


Fig. 2. The MINT Model $T(\nabla \mathbf{w} | \theta)$ is trained to determine whether given data d was part of the training data $\mathcal{D}$ used to develop a Language Model M . The inputs for the MINT Model consist of gradients $\nabla \mathbf{w}$, specifically the gradients of the model's parameters with respect to the data samples d .

A. Problem Statement

(See Fig. 2.) Given a training dataset $\mathcal{D}$ and an external dataset $\mathcal{E}$, we define a set of samples d that belong to $\mathcal{D} \cup \mathcal{E}$. We assume the existence of a model M that has been trained on $\mathcal{D}$ for a specific task. For each input sample d , the model M produces an output y as a function of d and a set of parameters $\mathbf{w}$ learned during the training process:

$$y = M(d | \mathbf{w})$$

In this context, we propose that an authorized auditor has limited access to the trained model M . This access enables the auditor to gain insight into the behavior of the model as it processes data d . Specifically, the auditor can access the model gradients, denoted as $\nabla \mathbf{w}$, which are calculated as the data sample d passes through the trained model. These gradients provide a detailed view of how the model parameters respond to input data. Gradients can be extracted from each weight w_i in the network, i.e., from different layers at various depths of the model. This ability to access gradients from various layers allows the auditor to analyze the influence of different parts of the model architecture on the final output.

MINT aims to identify whether the given data instance d is part of the training dataset $\mathcal{D}$ used to train model M . To achieve this, an authorized entity uses training data $\mathcal{D}$ and external data $\mathcal{E}$ to train a MINT auditing model $T(\cdot | \theta)$. The MINT model is trained with the model gradients M , $\nabla \mathbf{w}$, calculated as training data $\mathcal{D}$, and external data $\mathcal{E}$ ($\mathcal{E} \notin \mathcal{D}$) are passed through M . This ensures that the auditing model captures the audited model's responses to both sets. The MINT model then predicts whether a data sample d belongs to training data $\mathcal{D}$ or external data $\mathcal{E}$. MINT models take advantage of the memorization capabilities inherent in machine learning processes. The fundamental elements of MINT are the following.

- **Audited Model M :** a trained Language Model, defined by a set of learned parameters $\mathbf{w}$.
- **Training Dataset $\mathcal{D}$ and External Dataset $\mathcal{E}$:** In this work $\mathcal{D}$ is a collection of text samples used to learn linguistic patterns, word relationships, and contextual meaning related to text classification tasks. The External Data $\mathcal{E}$ is a collection of text samples that is not included in the training dataset ($\mathcal{D}$).
- **Gradients $\nabla \mathbf{w}$:** a gradient refers to the vector of partial derivatives of a loss function with respect to the model's parameters (weights). Indicates the direction and magnitude of the steepest ascent of the loss function in the parameter space. During training, the gradients are computed via backpropagation and are used in gradient-based optimization algorithms (such as popular Stochastic Gradient Descent, Adam, or RMSprop) to update the model's parameters in the direction that minimizes the loss. Gradients provide crucial information about how each parameter should be adjusted to reduce prediction error and improve learning. The gradients of the model weights $\mathbf{w}$ were calculated with respect to the input samples d . The hypothesis underlying this work is that the gradients of samples $\mathcal{D}$ will exhibit patterns different from those of the gradients obtained from samples $\mathcal{E}$.
- **MINT Model $T(\nabla \mathbf{w} | \theta)$ (Auditing Tool):** a model trained to perform the membership inference task, using input gradients to predict whether a given sample d belongs to the training dataset $\mathcal{D}$ or to external data $\mathcal{E}$. The architecture of the MINT model is a fully connected neural network with 3 layers, consisting of dense layers with 256, 128, and 64 neurons (ReLU), followed by a sigmoid output layer with one unit. The input size of the MINT model depends on the number of gradients of the audited model. In order to reduce the size of the input space, in our experiments we used the gradients of the last three layers of the audited model (binary cross-entropy as the loss function, Adam optimization). Training runs for 100 epochs with a batch size of 64.

Accuracy of Audited Language Models for Different Text Classification Benchmarks							
Databases	BLSTM	BERT	DistilBERT	XLNet	XLNet-Large	ELECTRA	ELECTRA-Large
AgNews (177 tokens)	91.17%	91.30%	93.79%	88.98%	91.54%	88.09%	92.73%
DBPedia (172 tokens)	96.54%	97.45%	97.55%	89.46%	92.45%	90.78%	91.25%
IMDB (165 tokens)	86.34%	90.16%	89.30%	88.21%	90.17%	84.24%	87.52%
SteamReviews (50 tokens)	85.67%	86.11%	86.29%	77.35%	81.46%	81.36%	83.45%
Sentiment140 (170 tokens)	76.19%	78.23%	79.25%	71.96%	72.45%	74.44%	73.24%
TwitterSentiment (165 tokens)	86.82%	78.37%	84.99%	74.62%	79.41%	70.23%	72.81%

TABLE I

CLASSIFICATION ACCURACY OF THE AUDITED LANGUAGE MODELS EVALUATED ON MULTIPLE TEXT CLASSIFICATION BENCHMARKS (MAXIMUM NUMBER OF TOKENS PER SAMPLE IN BRACKETS). CLASSIFICATION ACCURACY (IN PERCENTAGES) ACHIEVED BY EACH MODEL: BLSTM, BERT, DISTILBERT, XLNET, XLNET-LARGE, ELECTRA, AND ELECTRA-LARGE.

IV. EXPERIMENTAL FRAMEWORK

We begin by introducing the text databases used in this study, which consist of six different datasets representing three distinct NLP tasks: one for Topic Classification, one for Entity Linking, and four for Sentiment Analysis. We will provide an overview of their content and structure. Following this, we will describe the audited Language Models, detailing their characteristics and configurations. Finally, we will outline the experimental protocol.

A. Databases

- **AGNews**. [33] A data set for topic classification containing 127,600 news articles categorized into four topics: world, sports, business, and science/technology.
- **DBPedia**. [34] An Entity Linking dataset derived from *Wikipedia*, comprising 342,782 articles classified into 70 hierarchical categories.
- **IMDB**. [35] A text classification resource for sentiment analysis, including 50,000 movie reviews, equally divided between positive and negative sentiments.
- **Steam Reviews**. [36] A sentiment analysis dataset containing 392,200 reviews from *Steam* platform users, each review labeled as positive or negative.
- **Sentiment140**. [37] A dataset used for sentiment analysis, particularly focused on classifying tweets into positive, negative, or neutral sentiments, with 1.6 million labeled tweets.
- **TwitterSentiment**. [38] A sentiment analysis dataset consisting of 76,400 tweets, categorized into one of four labels: positive, negative, neutral, or irrelevant.

B. Audited Language Models

Different models M are evaluated to analyze whether variations in depth and complexity of the audited model affect MINT performance. The Language Models evaluated in this study were selected for their diversity in architectures, sizes, and complexities, which provides a comprehensive comparison across different types of models. These include **BERT** [39] and **DistilBERT** [40], which represent widely used encoder-based architectures; **XLNet** and **XLNet-Large** [41], which incorporate both autoregressive and bidirectional elements; **ELECTRA** and **ELECTRA-Large** [42], known for their more efficient pretraining techniques; and a **Custom BLSTM**, included to explore the performance of recurrent

architectures compared to transformers. This selection aims to assess the effects of different model characteristics, such as architecture type, model size, and training complexity, on MINT performance.

C. Experimental Protocol

To evaluate the MINT model in various scenarios, the models listed in Section IV-B were trained using the databases outlined in Section IV-A. Each language model utilizes its own tokenizer throughout the process. All language models have been trained for a total of 50 epochs with a batch size of 512, using Adam as the optimizer. Data separation into training and test sets was carried out assigning 65% to training and 35% to tests. The results of the trained models for the different tasks are presented in Table I.

Once the models were trained, we obtained the gradients of a subset of samples from the dataset, including the fragments $\mathcal{D}$ and $\mathcal{E}$. The goal is to make the training subset $\mathcal{D}$ as small as possible to minimize the number of samples that the developer of the audited model must provide. Consequently, the number of samples is gradually reduced throughout the process (as we will see in the next sections).

This approach is applied in two experimental setups: **Intra-Database Evaluation (1-vs-1)** and **Mixed-Database Evaluation (1-vs-6)**, as described in Section 5. In the Intra-Database Evaluation setup (1-vs-1), gradients are extracted directly from the $\mathcal{D}$ and $\mathcal{E}$ splits of the same dataset used during training, with a focus on comparing the dataset to itself. In this experimental setup, training samples from $\mathcal{D}$ and $\mathcal{E}$ consist of text from the same domain. For the Mixed-Database Evaluation (1-vs-6), gradients $\mathcal{E}$ are extracted not only from unseen samples of the original dataset but also from external text sources, simulated using data from other databases with distinct thematic domains. These additional data are incorporated to better replicate real-world scenarios.

The selected samples, which include both $\mathcal{D}$ and $\mathcal{E}$ samples, are classified based on two classes. This classification is then used as input for the MINT model, which predicts whether a sample d belongs to $\mathcal{D}$ or $\mathcal{E}$. By extracting and analyzing gradients, this process identifies patterns in the behavior of the model, providing a robust framework for auditing machine learning models and enhancing transparency. After extraction, the gradients are flattened and

those corresponding to the last three layers are selected to be used as input for the MINT model. The MINT model itself is trained for a total of 100 epochs with a batch size of 32 for all models. Data separation is carried out by assigning 65% for training and 35% for testing.

V. EXPERIMENTS AND RESULTS

A. Intra-Database Evaluation

In the Intra-Database evaluation configuration (1-vs-1), both $\mathcal{D}$ and $\mathcal{E}$ are from the same dataset to assess the model's performance on data with similar characteristics, such as the text domain and sources. Using the same dataset ensures that the samples come from a similar context and present a common distribution. This approach removes potential latent variables that could arise if data from different sources or with varying features were used, making the comparison more consistent. This setup helps to guaranty that the results are not influenced by external factors or inconsistencies, allowing for a clearer understanding of how well the model can classify and generalize within the given dataset. The results of the Intra-Database evaluation are presented in Table II.

The results in Table II show that the model effectively distinguishes between a sample d that belongs to $\mathcal{D}$ and a sample d that belongs to $\mathcal{E}$ that has the same context and distribution, highlighting its strong generalization within the dataset. The AUC values in the tables demonstrate robust performance under most conditions. High AUC scores indicate the model's ability to reliably differentiate between classes, reflecting its accuracy with sufficient data. As expected, the best results for all models are obtained when the number of samples to train the MINT Model T is higher. In contrast, as the number of MINT training samples decreases, the AUC gradually decreases, reaching its minimum value when the samples are 750 each (750 samples from $\mathcal{D}$ and 750 samples from $\mathcal{E}$). In most of the experiments, 3,000 training samples (1,500 from each class $\mathcal{D}$ and $\mathcal{E}$), was enough to obtain very competitive performances.

The limited sample size likely prevents an adequate representation of the data, impedes the learning and generalization of the model, and its ability to distinguish between samples belonging to $\mathcal{E}$ and samples belonging to $\mathcal{D}$.

B. Mixed-Database Evaluation

In the Mixed-Database evaluation setup (1-vs-6), our objective is to assess the generalization capability of the MINT model by simulating a realistic scenario where training and external data come from different sources. Specifically, we select one dataset to serve as training data $\mathcal{D}$, and use the remaining five datasets—along with unseen samples from $\mathcal{D}$ itself—as external data $\mathcal{E}$. These external sources introduce domain shifts and varying content, allowing us to evaluate how well the model can distinguish between samples it has seen during training and those it has not, across diverse distributions. By introducing these diverse sources, the evaluation reflects real-world scenarios and how the model adapts to different data distributions and contexts,

Databases	Number of MINT Samples ($\mathcal{D}$ / $\mathcal{E}$)				
	2.5K/2.5K	2.25K/2.25K	1.5K/1.5K	1.25K/1.25K	750/750
BLSTM (13M parameters)					
AgNews	0.9224	0.9199	0.8702	0.8051	0.7212
DBPedia	0.9257	0.9261	0.8560	0.7795	0.7353
IMDB	0.9352	0.9142	0.8942	0.8513	0.8120
Sentiment140	0.9273	0.9017	0.8252	0.7974	0.7061
TwitterSentiment	0.9482	0.9351	0.8990	0.7983	0.7628
SteamReviews	0.9355	0.9271	0.8479	0.8113	0.7596
BERT (110M parameters)					
AgNews	0.9566	0.9378	0.8841	0.8591	0.7393
DBPedia	1.0000	1.0000	0.9982	0.9979	0.9918
IMDB	0.9956	0.9982	0.9906	0.9729	0.9614
Sentiment140	1.0000	1.0000	0.9999	0.9997	0.9965
TwitterSentiment	0.9996	1.0000	0.9937	0.9996	0.9893
SteamReviews	0.9921	0.9909	0.9527	0.9489	0.8993
DistilBERT (66M parameters)					
AgNews	0.9851	0.9797	0.9600	0.9101	0.8343
DBPedia	0.9994	0.9912	0.9934	0.9882	0.9355
IMDB	0.9575	0.9793	0.9372	0.8597	0.8269
Sentiment140	0.9561	0.9225	0.8912	0.8640	0.7546
TwitterSentiment	0.9968	0.9974	0.9147	0.8641	0.8544
SteamReviews	0.9824	0.9892	0.9668	0.9404	0.8757
ELECTRA (110M parameters)					
AgNews	1.0000	1.0000	0.9960	0.9992	0.9945
DBPedia	0.9983	0.9997	0.9965	0.9999	0.9955
IMDB	0.9626	0.9802	0.9525	0.9532	0.8752
Sentiment140	1.0000	1.0000	0.9995	0.9979	0.9818
TwitterSentiment	1.0000	1.0000	0.9988	0.9988	0.9908
SteamReviews	0.9993	0.9991	0.9976	0.9875	0.9618
ELECTRA-Large (335M parameters)					
AgNews	0.9991	0.9954	1.0000	0.9852	0.9751
DBPedia	0.9932	0.9997	0.9974	1.0000	0.9942
IMDB	1.0000	1.0000	0.9912	0.9923	0.9823
Sentiment140	1.0000	1.0000	1.0000	0.9998	0.9900
TwitterSentiment	0.99975	0.9975	1.0000	0.9995	0.9895
SteamReviews	1.0000	1.0000	1.0000	0.9944	0.9884
XLNet (110M parameters)					
AgNews	1.0000	0.9975	1.0000	0.9984	0.9541
DBPedia	0.9987	1.0000	1.0000	1.0000	0.9545
IMDB	0.9941	0.9998	0.9954	0.9754	0.9432
Sentiment140	0.9952	0.9985	0.9852	0.9621	0.9760
TwitterSentiment	1.0000	1.0000	0.9963	0.9900	0.9569
SteamReviews	1.0000	1.0000	0.9952	0.9754	0.9784
XLNet-Large (340M parameters)					
AgNews	0.9993	1.0000	0.9824	0.9952	0.9421
DBPedia	1.0000	0.9999	1.0000	0.9870	0.9447
IMDB	0.9977	0.9903	0.9814	0.9710	0.9554
Sentiment140	0.9982	0.9945	0.9964	0.9927	0.9852
TwitterSentiment	0.9997	0.9984	0.9856	0.9875	0.9510
SteamReviews	1.0000	1.0000	1.0000	0.9998	0.9935

TABLE II
MINT AUC SCORES FOR DIFFERENT DATASETS AND $\mathcal{D}$ AND $\mathcal{E}$ SIZES
ACROSS THE LANGUAGE MODELS EVALUATED FOR THE
INTRA-DATABASE SETUP (1-VS-1).

providing insight into its ability to generalize beyond the specific nature of $\mathcal{D}$, as shown in Table III.

The results follow the same trend as those obtained in Section V-A. It can be seen from the high AUC values that the MINT model is able to differentiate between $\mathcal{D}$ and $\mathcal{E}$ from the same database, as well as from different sources.

All experiments were carried out on a PC with Intel Core i9-14900 5.8 GHz, 64GB RAM, and 2× Nvidia RTX4090 (24GB VRAM). The proposed MINT models can be trained with very limited computational resources, and they can be considered medium-light models.

Databases	Number of MINT Samples ($\mathcal{D}$ / $\mathcal{E}$)				
	2.5K/2.5K	2.25K/2.25K	1.5K/1.5K	1.25K/1.25K	750/750
BLSTM (13M parameters)					
AgNews	0.9335	0.9056	0.8852	0.8521	0.8107
DBPedia	0.9242	0.9123	0.8924	0.8856	0.8700
IMDB	0.9245	0.9256	0.8879	0.8421	0.7996
Sentiment140	0.9215	0.9352	0.8702	0.8615	0.8120
TwitterSentiment	0.9321	0.9351	0.9120	0.8623	0.8550
SteamReviews	0.9451	0.9387	0.8975	0.8569	0.7950
BERT (110M parameters)					
AgNews	0.9697	0.9716	0.9697	0.9510	0.9333
DBPedia	0.9996	1.0000	1.0000	0.9999	0.9997
IMDB	0.9966	0.9945	0.9797	0.9775	0.9560
Sentiment140	0.9994	1.0000	0.9994	0.9968	0.9910
TwitterSentiment	0.9993	0.9987	0.9986	0.9930	0.9734
SteamReviews	0.9880	0.9858	0.9598	0.9519	0.9097
DistilBERT (66M parameters)					
AgNews	1.0000	0.9999	0.9990	0.9998	0.9919
DBPedia	0.9989	0.9998	0.9989	0.9990	0.9996
IMDB	0.9779	0.9320	0.9020	0.9268	0.8639
Sentiment140	0.9381	0.9203	0.8897	0.8241	0.8196
TwitterSentiment	0.9982	0.9988	0.9943	0.9938	0.9858
SteamReviews	0.9498	0.9831	0.9350	0.9481	0.9050
ELECTRA (110M parameters)					
AgNews	0.9987	0.9972	0.9871	0.9821	0.9721
DBPedia	0.9989	0.9998	0.9989	0.9996	0.9998
IMDB	0.9970	0.9960	0.9943	0.9940	0.9786
Sentiment140	0.9997	0.9992	0.9976	0.9978	0.9844
TwitterSentiment	0.9998	0.9998	0.9994	0.9985	0.9974
SteamReviews	0.9983	0.9978	0.9886	0.9847	0.9647
ELECTRA-Large (335M parameters)					
AgNews	1.0000	1.0000	0.9871	0.9821	0.9721
DBPedia	1.0000	1.0000	1.0000	1.0000	0.9998
IMDB	0.9970	0.9960	0.9943	0.9940	0.9786
Sentiment140	1.0000	1.0000	0.9906	0.9964	0.9901
TwitterSentiment	1.0000	1.0000	1.0000	0.9992	0.9984
SteamReviews	1.0000	0.9995	0.9977	0.9901	0.9896
XLNet (110M parameters)					
AgNews	0.9956	0.9954	0.9940	0.9967	0.9359
DBPedia	0.9997	0.9993	0.9988	0.9986	0.9854
IMDB	0.9814	0.9966	0.9821	0.9662	0.9418
Sentiment140	0.9979	0.9929	0.9778	0.9740	0.9715
TwitterSentiment	1.0000	0.9912	1.0000	0.9972	0.9994
SteamReviews	0.9998	0.9999	0.9991	0.9980	0.9820
XLNet-Large (340M parameters)					
AgNews	1.0000	1.0000	1.0000	0.9966	0.9456
DBPedia	1.0000	0.9993	1.0000	0.9986	0.9854
IMDB	0.9975	1.0000	0.9954	0.974	0.961
Sentiment140	0.9984	0.9973	0.9954	0.9841	0.9815
TwitterSentiment	1.0000	0.9915	1.0000	0.9947	0.9965
SteamReviews	1.0000	1.0000	1.0000	0.9972	0.9920

TABLE III

MINT AUC SCORES FOR DIFFERENT DATASETS AND $\mathcal{D}$ AND $\mathcal{E}$ SIZES
ACROSS THE LANGUAGE MODELS EVALUATED FOR THE
MIXED-DATABASE SETUP (1-VS-6).

VI. CONCLUSIONS

This work demonstrates the feasibility and importance of auditing training data in modern language models. By adapting and extending the MINT method to the textual domain, we show that it is possible to determine whether a specific text was included in a model training, supporting greater transparency and accountability in AI development.

We evaluated the approach in six datasets and seven Transformer-based language models. The experimental results show strong performance, with AUC scores ranging from 70% to 99%, depending on the size of the data set, the complexity of the model, and the number of samples

provided to MINT. Importantly, the effectiveness of auditing depends on striking the right balance in the number of query samples: too few can reduce accuracy, while too many may yield better results but require a high volume of queries to the audited model. Achieving this balance is the key to ensuring reliable and realistic audit performance

The study further reveals that the attributes of the model, such as architecture, scale and training dynamics, have a direct influence on auditability. A clear correlation between model complexity and gMINT performance highlights the sensitivity of the method to memorization patterns, which is essential for effective auditing. Beyond extending MINT to the language domain, this work lays the foundations for broader applications of auditing tools in artificial intelligence. By enabling systematic detection of the inclusion of training data, our method supports the development of more fairer, transparent, and regulation-compliant NLP systems.

Future progress in this direction can help establish standardized practices for data auditing, contributing to a more responsible and trustworthy AI ecosystem.

VII. LIMITATIONS AND FUTURE WORK

- Our experiments focus on text classification tasks. Future research should explore the application of MINT to language models trained for other tasks, particularly increasingly popular generative models. Generative models are characterized by an extremely high number of parameters (in the billions) and a training process that involves vast amounts of data. Our experiments initially suggest that a larger number of parameters help to improve the detection of training data.
- Our experiments assume a collaborative or, at the very least, non-intrusive stance from the owner of the audited model. Our method requires access to the model's gradients, which are not available in proprietary language models. Future work should investigate vulnerabilities and develop countermeasures for scenarios in which the model owner actively seeks to obfuscate the training data (e.g., introducing small variations in the training data).
- Due to space limitations, we have not included experiments that analyze how the number of tokens affects the performance of MINT models. The experiments presented in Sect. V cover a range of token lengths, varying from 50 (SteamReview) to 177 (AgNews). Future research should further investigate the impact of token count on performance, especially in cases where the number of tokens is low.

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